Local Factor Rigidity and Mass–Defect Bounds for Langton's Ant

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Abstract

The cyclic ten-letter factors of the standard Langton highway impose both asymptotic rigidity and a finite-time mass constraint. We prove that every physical ten-legal infinite stream eventually has period 104, 28 or 24. For finite initial black mass m, only the standard highway remains and its first entry time is at most $12m+212$, with optimal linear coefficient 12. A bounded-fold graph also computes exact first violation times and complete compressed endpoints in polynomial bit time, with quadratic finite-event deadlines. Independently, a finite-word potential certificate proves $3(t-s)-26(|B_t|-|B_s|) \le 351N_{10}(s,t)+322$ on every trajectory interval. The additive constant and the standard mass-growth slope are sharp. Exact finite kernels accompany the all-parameter physical arguments.

1 Introduction

The standard Langton highway has a long period but a short local signature. Its cyclic ten-letter factors already constrain every physically realizable infinite turn stream to three eventual modes. Finite black support selects one of them, the standard period-104 highway. The same local signature also controls finite pieces of arbitrary trajectories through a quantitative balance between black-cell growth and forbidden windows.

This paper develops these two consequences together. The first is an attraction theorem under a future-language hypothesis, with a sharp linear dependence on initial black mass. It yields a polynomial-time decision procedure for the existence of a particular local violation event. The second is an unconditional inequality on every finite orbit interval. Its proof uses a separate graph of finite legal words, so it does not assume that a finite fragment can continue legally forever.

Gajardo's symbolic projection and forbidden-word framework [1] provide the finite-word viewpoint. Gajardo, Lutfalla and Rao [2] discuss highway certification; Jillhewar's full-drift chronology [3] supplies the periodic realization test used here. Our contributions are the concrete ten-factor classification, the all-parameter exclusion in its nonsimple recurrent component, the sharp transient coefficient, the local-target algorithm, and the mass–defect inequality.

2 Model and principal results

A state consists of the complete black set B, a position in $\mathbb{Z}^2$, and a cardinal heading. White turns right (R), black turns left (L); the cell then flips and the ant moves. Time denotes the state before the update. Let W be the standard 104-letter turn word in the appendix and let $\mathcal{L}_k$ contain all its cyclic factors of length k. A stream is k-legal if every future length-k window belongs to $\mathcal{L}_k$.

Theorem A (eventual modes). Every physical ten-legal stream on an arbitrary initial colouring eventually has least period 104, 28 or 24, with the respective words W, U, T given below.

Theorem B (finite-mass attraction). For a complete finite initial black set of cardinality m, a ten-legal future reaches the standard trace by

$$\tau(X) \le 12m + 212.$$

The coefficient 12 is optimal: a family with $m_n = 2n + 13$ has $\tau_n = 24n - 3$ for every $n \ge 1$. The intercept 212 is a proved upper bound.

Theorem C (arbitrary-colouring threshold). An explicit computable infinite colouring has a nine-legal stream that is not eventually periodic. Thus ten is the least factor length forcing eventual periodicity on arbitrary colourings. This sharpness statement uses infinite support.

Theorem D (local event). Let J_{10} test the true preceding nine turns and the current turn for membership outside $\mathcal{L}_{10}$, recovering the past by inverse dynamics. On every finite input, existence of a future J_{10} is decidable in polynomial bit time for sparse binary coordinates.

Theorem E (mass–defect balance). On every interval $[s, t)$ of every finite-support trajectory,

$$3(t - s) - 26\big(|B_t| - |B_s|\big) \le 351 N_{10}(s, t) + 322,$$

where N_{10} counts forbidden ten-letter windows wholly contained in the interval. The constant 322 is attained by a 90-step witness; the mass slope $3/26$ is asymptotically sharp. The penalty 351 is not asserted optimal.

Theorems A–D concern infinite legal futures or a decision using their contrapositive. Theorem E instead rests on a complete finite-word potential certificate. This separates their logical dependencies while using the same local language.

3 From local language to eventual attraction

3.1 Finite-word feasibility and graph coverage

For any finite prescribed turn word, the geometric path is fixed after choosing an initial pose. It is physically realizable if and only if consecutive reads of each same physical cell have alternating letters. Necessity is the flip rule. For sufficiency, make a visited cell initially black exactly when its first requested letter is L; leave every unvisited cell white. Induction over the finite word proves execution. This is established methodology, not a new criterion; see Gajardo (2003), Subsection ef{ten-3}.

Let E contain EVERY physically feasible 150-letter word whose 10-factors lie in $\mathcal{L}_{10}$. Construct a graph G with edge $v_0 \cdots v_{148} \to v_1 \cdots v_{149}$ for each $v \in E$. No word is removed because it looks unlikely or because its seed is large. Complete enumeration follows all whitelist-compatible extensions and rejects exactly the same-cell alternation conflicts. Induction on word length proves completeness of that enumeration.

Any actual infinite 10-legal stream gives an infinite path in G, since each of its length-150 windows is feasible under its own actual execution. The graph can admit paths that fail longer physical consistency. It is an OVERAPPROXIMATION; this does not invalidate exclusion, but prevents interpreting every graph path as a physical orbit.

The immutable certificate `verification/ten/certificates/windows10.txt` contains 6,605 edge words. Both verifiers regenerate the entire set and require exact equality, including cardinality and absence of duplicates. Their complete graph has 6,501 vertices and precisely three cyclic strongly connected components:

Component	Vertices	Internal edges	Structure
$\mathcal{W}$	104	104	exact standard simple cycle
$\mathcal{T}$	24	24	exact T simple cycle
$\mathcal{H}$	366	370	not a simple cycle

There are no other cyclic components, including self-loops. The first two component edge sets equal all cyclic length-150 factors of W and T. Strongly connected components and reachability are checked on the full graph, not only on a pruned experimental sample.

The non-simple component is crucial: ordinary finite-state periodicity reasoning cannot dispose of it. It is dealt with next by an all-length physical conflict, rather than by assuming its paths are periodic.

3.2 Exact two-loop description of the non-simple component

The certificate `quotient.tsv` assigns each of the 366 vertices of $\mathcal{H}$ a state in $\{0, \ldots, 101\}$. It is a label-preserving homomorphism onto this explicit graph:

- for each $0 \le i < 102$, the edge labelled B_i leads from i to $i+1$ modulo 102;
- one additional edge, labelled L, leads from 0 to 75.

Here B is the 102-letter word in the appendix, $B_0 = \mathrm{R}$, and the alternative short loop is $A = \mathrm{L}B_{75} \cdots B_{101}$, of length 28. Every one of the 370 internal edges is checked against this formula. The homomorphism is used only in the forward direction: every infinite internal path projects to a word that, after at most 101 letters, is an arbitrary concatenation of A and B.

The edge label is the letter APPENDED to the 149-letter history. It is shifted 149 places from the original stream. This finite shift is harmless for eventual-mode classification. Subsection ef{ten-5} instead uses the CURRENT letter v_0 of a future-window vertex, and does not confuse these conventions.

The exact geometric data, starting north at the origin, are

$$A : (0, -4),\ h = 0; \qquad B : (1, 11),\ h = 2.$$

Thus A translates and B reverses the heading. The A word is a cyclic phase of U.

3.3 Three foldbacks are impossible, at every pair of lengths

Lemma 1. For every pair of integers $n, m \geq 0$, the finite word BA^nBA^mB is physically infeasible, independently of its initial colouring.

In each case below there are two consecutive visits to a cell q both requesting R:

Parameter range	q	Earlier time	Later time
$n = 0$	$(0, 2)$	32	184
$n \geq 1,\ m \geq n$	$(0, 2)$	32	$184 + 56n$
$n \geq 1,\ 0 \leq m < n$, $r = n - m$	$(1, 4r)$	$25 + 28r$	$205 + 28(n + m)$

All times are pre-update indices in the displayed finite word. It suffices to verify their consecutiveness, not that the rest of the candidate word is feasible.

Proof. Let the five block boundaries start at times 0, 102, $102 + 28n$, $204 + 28n$, and $204 + 28(n + m)$. Their respective poses are $(0, 0), N$; $(1, 11), S$; $(1, 11 + 4n), S$; $(0, 4n), N$; and $(0, 4(n - m)), N$.

The following finite coordinate facts are computed directly from the two displayed words and checked independently: A has $-3 \leq x \leq 0$; on its column $x = 0$ the pairs (index, height, letter) are

$$(0, 0, L), (7, -1, R), (8, -2, R), (15, -3, L), (16, -2, L), (27, -5, L).$$

In B, the only visit to $(0, 2)$ is index 32, R. All B visits at $x = 0$ have $y \leq 2$. Visits to $(1, 9)$ are indices 82,R and 90,L; visits to $(1, 4)$ are 45,L and 53,R; visits to $(1, 8)$ are 73,L and 81,R. On $x = 1$ its maximum height is 12.

If $n = 0$, the reflected second B reads $(0, 2)$ at its local $(1, 9)$, index 82. The previous visit is the first B's index 32. This gives the first row.

If $n \geq 1$, the first A run lies at $x \geq 1$, so cannot revisit $(0, 2)$. In the second B, a visit there would require local $(1, 9 + 4n)$, above every $x = 1$ visit. In the second A run, the first matching copy is number $n - 1$ (zero-based), at local index 8, with letter R. All earlier copies miss: on $x = 0$ their required local height is below -5. The visit time is $204 + 28n + 28(n - 1) + 8 = 184 + 56n$. This copy exists when $m \geq n$.

Finally let $r = n - m > 0$. For $r = 1$ or 2, the last earlier visit to $(1, 4r)$ is the first B's index 53 or 81, both R. The first A run lies at heights at least 11 and cannot reach those two cells. For $r \geq 3$, the last visit in the first A run is copy $r - 3$, local index 7, R, at time $102 + 28(r - 3) + 7 = 25 + 28r$. The column-$x = 0$ list for A shows no later visit to that cell in the run; any visit in the first B precedes it. A visit by the second B would require local $(0, 11 + 4m)$, impossible since its column-$x = 0$ maximum is 2.

The second A run lies at $x \le 0$. The third B, based at $(0, 4r)$, first reaches $(1, 4r)$ at index 1, R. This proves the third row and the lemma. □

An infinite concatenation of A and B with infinitely many B's contains a forbidden three-B subword. Hence it has only finitely many B's and is ultimately A^ω. This argument disposes of arbitrary interleavings and unbounded run lengths, not only tested values of n, m.

An infinite path in a finite graph eventually remains in one cyclic strongly connected component. In $\mathcal{W}$ and $\mathcal{T}$ it is forced periodic by the simple cycles. In $\mathcal{H}$ the preceding homomorphism and Lemma 1 force an A-periodic suffix. This proves Theorem A.

3.4 Finite support and the transient bound

For a heading-resetting periodic word of length P with nonzero drift d, partition its phase positions by differences in $\mathbb{Z}d$. For a class write $q_i = b + a_i d$. Visits to $b + nd$ are eligible when $a_i \le n$, and their chronological order is decreasing a_i, then increasing i. A cell's required initial colour is the first letter of that ordered eligible list. This is the full-drift chronology method of Jillhewar (2026), Theorem 3.1 and Remark 3.2.

The words U and T pass all per-cell alternation tests but have respectively five and two stabilized classes starting L. Each such class demands infinitely many DISTINCT initially black cells. Neither word can be executed forever from finite support. Any finite prelude preserves finite support, so these modes also cannot occur as eventual tails of a finite-support orbit. Consequently every finite-support infinite 10-legal path eventually enters $\mathcal{W}$.

The complete graph gives 925 vertices that can reach $\mathcal{W}$. They exclude $\mathcal{H}$ entirely and include $\mathcal{T}$. Apart from $\mathcal{W}$ and $\mathcal{T}$, they are acyclic. A path cannot leave a strongly connected component and later return to it. The current letters of a run of L successive vertices in $\mathcal{T}$ form a length-L prefix of a phase of T^ω.

Let $b_j(L)$ count distinct cells whose FIRST read is L in this prefix starting at phase j. If the complete black mass at the run's start is m_T, physical execution requires $b_j(L) \le m_T$. The count includes only first reads relative to that run; cells written during the run cannot pay for them.

Lemma 2. For all phases j and all integers $L \ge 0$,

$$L - 12b_j(L) \le 5.$$

Proof. Chronological classes give the first-L read times as a finite set F_j and two arithmetic rays $r_{j,1} + 24\mathbb{N}$, $r_{j,2} + 24\mathbb{N}$. Their exact table is in the appendix. The sets count distinct physical cells, not repeated L reads. Put $D_j = \max(\{0\} \cup \{f + 1 : f \in F_j\} \cup \{r_{j,1}, r_{j,2}\})$. For every $L \ge D_j$, directly from these rays,

$$b_j(L + 24) = b_j(L) + 2.$$

Thus $L - 12b_j(L)$ is 24-periodic for all $L \geq D_j$. Checking the finite interval $0 \leq L < D_j + 24$ covers all lengths. Both implementations derive the class table independently and verify that the maximum across all phases is 5. This is an exact finite reduction with its all-length periodicity proof, not a finite-prefix extrapolation. □

Before its first $\mathcal{T}$ vertex a path uses only the 54 acyclic strict ancestors of $\mathcal{T}$. Give such a vertex v weight

$$c(v) = 1 + 12\delta(v_0), \qquad \delta(R) = 1,\ \delta(L) = -1.$$

Complete acyclic dynamic programming gives maximum total weight 70 for a path ending at $\mathcal{T}$. Therefore if t_{pre} updates precede that run and their net black change is Δ, then $t_{pre} + 12\Delta \leq 70$ and $m_T = m + \Delta$. After departure from $\mathcal{T}$, the maximum acyclic path length before $\mathcal{W}$ is 137. The exit-producing update from a $\mathcal{T}$ vertex is included in its T-prefix, not omitted. By Lemma 2,

$$\tau \leq t_{pre} + (12m_T + 5) + 137 \leq 12m + 70 + 5 + 137 = 12m + 212.$$

Paths avoiding $\mathcal{T}$ altogether have length at most 167 before $\mathcal{W}$, also below this bound. A path whose initial vertex is already in $\mathcal{W}$ has zero such entrance time. Once the path first reaches $\mathcal{W}$, leaving it would preclude a later return; because its eventual component must be $\mathcal{W}$, it never leaves. Its future is standard from that time.

These are future-window vertices at times starting at zero, so the transient bound needs no additional window warm-up. Every complete initial black cell remains in the physical state; only its count is used in this inequality. Theorem B's upper bound follows.

3.5 An explicit family attaining coefficient twelve

Define

$$E = \{(-2,-1),(-1,0)\},$$

$$\begin{aligned} D = \{&(-1,-4),(-1,-3),(0,-5),(0,-3),(0,-2),(1,-6),\\ &(1,-5),(1,-4),(1,-3),(2,-3),(2,-2)\}. \end{aligned}$$

$$R = \{(-2,0),(-1,-1),(-1,0),(0,-1),(1,-1),(1,0)\}.$$

For $n \geq 0$, put

$$B_n = E \ \cup\ \{(-2,-2j),(0,-2j-1) : 1 \leq j \leq n\} \ \cup\ (D + (0,-2n)).$$

These sets are disjoint. Start north at the origin with exactly B_n black and all other cells white. Then $|B_n| = 2n + 13$.

The word T has read footprint within $[-2, 1] \times [-3, 0]$, net displacement $(0, -2)$, and resets heading. For every $n \geq 1$ its required initial black cells in that footprint are precisely $E \cup \{(-2, -2), (0, -3)\}$. All later rail cells and all cap cells are outside the footprint. The complete 24-step parity of visits, followed by set cancellation, gives the identity

$$F^{24}(B_n, 0, N) = \big((B_{n-1} + (0, -2)) \,\dot\cup\, R,\ (0, -2), N\big).$$

This is an identity of FULL states, not a core-only recurrence. A direct all-parameter derivation is: translate the rail terms with $j = 1, \ldots, n - 1$ by -2; they become the original terms $j = 2, \ldots, n$. The cap becomes $D + (0, -2n)$. Only the fixed finite set E, the first rail pair, and the translated E remain to compare, and their difference with R equals the exact net toggle set of T. This finite equality is checked. It also covers $n = 1$ where the shorter rail is empty.

The residual R is at heights -1 and 0. Every future repeated T segment based at height -2 or less remains strictly below it. The $n = 0$ complete state is the clean seed of the standard phase Z in the appendix; its full infinite execution is verified by chronology and an independent epoch-breakpoint formula. Its phase footprint has maximum height zero and drift $(-2, -2)$, so its continuation from height -2 also stays below R. Induction therefore retains and guards ALL successive residuals and proves the exact infinite word

$$T^n Z^\omega.$$

The factors of $T^2 Z^2$ verify every possible length-10 junction; T^ω and Z^ω are themselves 10-legal. The last three letters of T and Z agree, but the last four do not. Since Z has least period 104, any earlier standard future agreeing with Z^ω must be its unique phase-aligned left extension. Hence the first standard time is

$$\tau_n = \max(0, 24n - 3).$$

For $n \geq 1$, $\tau_n = 12|B_n| - 159$. An affine upper bound with slope less than 12 and fixed intercept fails for large n. This proves the sharp coefficient, without claiming 212 is sharp. Every member of this family is H-positive.

3.6 Nine factors admit an aperiodic infinite-background spiral

Let U and C be the 28- and 89-letter words in the appendix and let $n_i = 5 \cdot 2^i$. Consider

$$a = \prod_{i=0}^{\infty} U^{n_i} C.$$

The prescribed geometry is explicitly computable, starting at the origin heading north. Its all-future physical feasibility follows from the next separation proof; it is not assumed.

A U block restores heading, moves $(0, -4)$, and has footprint in $[0, 3] \times [-5, 0]$. A C block rotates the heading counterclockwise, moves $(13, -4)$, and has footprint in $[0, 14] \times [-4, 0]$. The finite word UCU is feasible. Every length-9 factor of U^2CU^2 is in $\mathcal{L}_9$.

The start heading of macro i is $-i$ modulo four. For $S = 16^k$, the four consecutive start positions are

$$p_{4k} = (-8S + 8, 4S - 4), \quad p_{4k+1} = (-8S + 21, -16S - 8),$$
$$p_{4k+2} = (32S + 25, -16S + 5), \quad p_{4k+3} = (32S + 12, 64S + 9).$$

These follow by the two block displacements; one group of four returns the heading and replaces S by $16S$. Entire macro read footprints lie in the respective rectangles

Index	x interval	y interval
$4k$	$[-8S + 8, -8S + 22]$	$[-16S - 8, 4S - 4]$
$4k + 1$	$[-8S + 21, 32S + 25]$	$[-16S - 8, -16S + 6]$
$4k + 2$	$[32S + 11, 32S + 25]$	$[-16S + 5, 64S + 9]$
$4k + 3$	$[-128S + 8, 32S + 12]$	$[64S - 5, 64S + 9]$

Each rectangle is disjoint from all nonadjacent earlier rectangles. For the first row, $k \geq 1$, earlier rectangles through $4k-2$ have $x \geq -S/2+8$, while the new maximum is $-8S+22$; use $15S > 28$. For the second row, earlier rectangles through $4k-1$ have $y \geq -S - 8$, while the new maximum is $-16S + 6$; use $15S > 14$. For the third row, earlier rectangles through $4k$ have $x \leq 2S + 25$, while the new minimum is $32S + 11$; use $30S > 14$. For the fourth row, earlier rectangles through $4k + 1$ have $y \leq 4S + 9$, while the new minimum is $64S - 5$; use $60S > 14$. The first two empty-earlier-set base cases are automatic. These inequalities hold for every $S = 16^k$ in the stated ranges.

It remains to control adjacent macros, which are not claimed disjoint. In local coordinates of the earlier macro, the next macro begins at $(13, -4n_i - 4)$ heading west. Its first U block can overlap the earlier C, but every later U block has $x \geq 17$ while C has $x \leq 14$; the next C is still farther right. The preceding U rail has $x \leq 3$, so is disjoint from the next macro. Within one macro, C can meet only the LAST U block: every earlier U block has minimum height at least $-4n_i + 3$, above C's maximum $-4n_i$. Thus all interactions with C are contained in the verified word UCU. Other repetitions are portions of the physically feasible U^ω trace. This proves alternating letters at every consecutive same-cell pair in the entire infinite word.

Colour a cell initially black precisely when its first prescribed read has letter L; leave unvisited cells white. The proved consistency makes this a genuine infinite classical-ant execution by induction over time. This colouring is computable: for a queried cell

of infinity-norm M, choose k with $16^k > M + 100$. Every rectangle from macro $4k$ onward lies outside $[-M, M]^2$ in at least one coordinate, so only the first $4k$ finite macros need be searched. It has infinite black support, since interior portions of arbitrarily long U rails demand fresh initial black cells.

The word $D_{bad} = \texttt{RLRRRRLRRL}$ is absent from $\mathcal{L}_{10}$ and occurs exactly once in each connector C, at offset 11. The finite check of U^2CU^2 excludes other occurrences at either junction, and U^ω has none. Successive occurrences are separated by $89 + 28n_{i+1}$, which grows without bound. An eventually periodic sequence with infinitely many occurrences of a fixed finite factor has bounded gaps between them. Hence a is not eventually periodic. This proves Theorem C, including its explicitly infinite-background boundary.

3.7 A radius-nine target decidable without waiting for a remote collision

The inverse of a literal step is unique: from post-pose (p, h) set $q = p - \mathrm{dir}(h)$, undo the flip at q, and undo the turn dictated by that recovered colour. Nine inverse steps read only cells within distance nine of the current head. Therefore the preceding-nine-turn test J_{10} is an intrinsic finite-local predicate, including at input time zero.

Let $M = 12|B_0| + 212$. The following total algorithm acts on an arbitrary finite input, with no legality promise:

1. Recover the true nine-turn past. Simulate up to time M, checking J_{10} before each update. Return YES with its first time if encountered.
2. At time M, test the complete-state predicate P_{now} for permanent standard P104. If it fails, return YES (a future event is forced by Theorem B). This is not an assertion that the event was observed or that its first time has been computed.
3. If P_{now} holds, simulate nine additional updates to check all windows crossing this boundary. Return YES if one is bad; otherwise return NO. All subsequent windows are inside the certified standard future.

Why is the second step sound? Absence of every future J_{10} implies legality of every forward length-10 window from time zero. Theorem B would force P_{now} by time M, contradicting the failed test. A far-away blocker cannot be omitted from that test.

The exact P_{now} test first simulates 104 turns on a copy. If they do not equal any phase of W, it returns false. Otherwise let K be the unique clean finite seed for that placed phase, computed by the chronological class formula. Let $q_i + \mathbb{N}d$ be its 104 future phase rays. The actual complete field realizes that future if and only if

$$(B \triangle K) \cap \bigcup_{i=0}^{103} (q_i + \mathbb{N}d) = \emptyset.$$

Sufficiency follows by first-difference induction; necessity follows at the first read of an initial differing cell. Testing finitely many complete-field points against 104 integer rays is exact, including arbitrarily distant markers. No repeated-prefix heuristic is used.

During the $O(m)$ ordinary updates the complete field has $O(m)$ cells, and coordinate bit lengths grow by only $O(\log(m+1))$ beyond their input sizes. Exact finite-set operations and integer ray tests therefore give a deterministic polynomial bit-time implementation (a balanced search tree can be used instead of hashing). The implementation uses Python integers and finite sets; even worst-case linear set lookups remain polynomial. There is no claim of a polynomial bound for the numerical first-event time or of a bound polynomial in the bit length of a COMPRESSED black count.

3.7.1 A fixed-mass late-event control

`verification/ten/certificates/far_marker.json` gives a complete 27-black standard state at a mature phase. Add one black marker at $(2n, 2n)$ for any $n \geq 20$. The new complete mass is always 28. The first contact is exactly at update $104n - 92$, and it changes a prescribed R into L, forming the forbidden ten-factor `RRRLLLLRLL`.

The all-parameter first-contact calculation is: for the supplied phase positions q_i, write $q_i = s_i(2, 2)$ whenever possible. The first visit to the marker has time $104n + \min_i(i - 104s_i)$. The minimum is -92, attained at phase 12, and all relevant epochs are nonnegative for $n \geq 20$. The marker is outside the finite true nine-step inverse history and the seed. Until contact, first-difference coupling preserves the entire standard path. The standard cyclic preceding nine letters and the changed L give exactly the stated bad factor.

For $n = 10^{12}$ the first event is at 103999999999908, but the new decision algorithm returns a forced-future YES after only 548 main-trajectory updates. The extra operations are nine inverse updates and the constant lookahead and ray tests, not simulation to the event. This family illustrates the distinction between reachability decision and event-time observation. Its coordinates encode the long waiting time.

3.8 Finitely many violations and standard convergence

For finite inputs, a standard future has no intrinsic violations after its first nine updates. Conversely, after the last violation every forward window is legal; Theorem B applied to that complete state gives a standard future. Hence

$$H(X) \iff \#\{t \geq 0 : J_{10}(F^t X)\} < \infty.$$

Theorem D decides each future-event query. This equivalence identifies the additional global assertion needed for convergence: finiteness of the event chain, rather than decidability of an individual query.

4 Exact first-violation macros

The preceding decision can be strengthened to return the first event itself and its complete post-event state. This requires a finite-path argument inside the nonsimple component, rather than an assumption of periodicity there.

Theorem F (exact event and finite prefixes). A complete finite field may be given as the XOR of M finite arithmetic segments whose strides belong to

$$\{(0,0),(\pm2,\pm2),(\pm2,0),(0,\pm2),(\pm4,0),(0,\pm4)\}.$$

Zero stride denotes a singleton. The first future J_{10}, or its permanent absence, is decidable in polynomial bit time in this representation. A finite answer returns its exact time and full state just after the event. One call uses at most five periodic plans and 1111 literal main-orbit updates, adding at most 1530 segment terms.

Let W_0 be the diameter in the maximum norm of the actual initial support and head. Conditional on n events existing, their post-event time satisfies

$$s_n \le n(212W_0 + 854199) + 415308n(n-1). \tag{MACRO1}$$

Every intermediate complete state lies in the initial convex hull enlarged by $[-1959n, 1959n]^2$. The first n events and their complete encoded endpoint are computable in poly$(M+n+b)$ bit operations, with n measured numerically and b the original coordinate and length bit size.

These macros preserve finite histories exactly. Their lift counts fold starts within one event-free traversal; a fresh event begins a fresh traversal. The following proof supplies both the finite graph coverage and the arithmetic continuation needed for this stronger conclusion.

4.1 Pinned base and finite-path coverage

The base contains every physically feasible 150-letter word all of whose ten-factors are in L10. Its 6605 edge words generate a graph G with 6501 vertices, the 149-letter prefixes/suffixes. Physical feasibility is exactly alternation of requested turns on consecutive reads of the same physical cell. This is the prior forbidden-word method, not claimed new here.

Every actual event-free continuation, once its 149-turn history is a G vertex, follows G. At most 149 nonviolating ordinary updates are needed to obtain such a vertex: afterwards the last 149 turns are wholly new, and if the next turn is legal their feasible 150-word is in the complete enumeration. If the next turn is illegal, it is returned as the event instead. This covers input time zero without inventing a periodic negative-time past.

The pinned graph has simple W and T components of lengths 104 and 24, and one component H with 366 vertices and 370 internal edges. The prior explicit quotient maps H to phases 0,...,101. There is an edge (j,B_j)->j+1 mod 102, plus (0,L)->75, where B is a 102-letter fold word beginning R. The short loop is A=L B_75...B_101, length 28. Their literal geometry is

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A: displacement (0,-4), heading restored;
B: displacement (1,11), heading reversed.
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Every internal H edge is checked against this quotient, not merely sampled. The already supplied all-parameter lemma is

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B A^n B A^m B is physically infeasible for ALL n,m>=0. (MACRO2)
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Its explicit consecutive same-cell/same-R conflicts and their parameter coverage are in Subsection 3.3. A finite word need not extend to an infinite legal word for (MACRO2) to apply. This is the essential finite-path rather than infinite-cycle input.

The base edge set and quotient have frozen SHA256 values in the source witness. The event runtime regenerates both and checks exact equality.

4.2 A complete bounded-fold lift

Outside H keep each vertex v with counter 0. Inside H use (v,c), c=0,1,2,3, except that (v,3) with quotient phase 0 is forbidden. On an internal H edge increase c exactly when its quotient starts the long loop, namely phase 0 with appended R. All other internal edges keep c. On entry into H start c=0; on exit forget c. Disallow c>3 and disallow arrival at phase 0 with c=3.

This permits two complete B loops and a third incomplete B loop; a third *completed* loop is excluded. An initial partial loop is not charged, so no assumption is made about the path before H entry. Between three long-loop starts at phase zero there are powers of A, and completing the third yields (MACRO2). Thus every real finite event-free path lifts. H cannot be revisited after leaving it because G's SCC condensation is acyclic. Counter reset on reentry cannot therefore mask additional physical folds.

The lift may still contain impossible paths. It is an exclusion-safe overapproximation, not a claim that every lifted path is physical.

Complete reconstruction yields 7595 vertices and 7763 edges. Its ONLY cyclic SCCs are simple cycles of lengths

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24, 104, 28, 28, 28.
```

The 28-cycles are A phases at counters 0,1,2. The 24- and 104-cycles are the prior T and standard words. No claim follows from cycle lengths alone; the words and their nonzero displacements are checked too.

Assign to each condensation component the three weights (one cyclic-component visit, one noncyclic-vertex update, period): respectively (1,0,P) on a cyclic component, and (0,1,0) on a noncyclic singleton. Backward longest-path values are nonnegative and, over every condensation edge C->D, satisfy

```
rank(C) >= weight(C)+rank(D)
```

coordinatewise. Terminal ranks dominate their own weights. Internal simple-cycle edges keep the same rank. All 7763 lifted edges are checked. The maxima of the three ranks are

```
5, 956, 212.                                                    (MACRO3)
```

Python uses the base's iterative two-pass SCC routine; the independent native checker reconstructs the lift and uses Tarjan SCCs. Sorted full edge and rank hashes agree. The source witness fixes the complete edge and rank hashes. Integer inequalities, complete reconstruction and (MACRO2), not a floating-point solver, carry the proof.

4.3 Exact physical acceleration on a simple component

At a cyclic vertex propose its internal periodic word u, of length P and nonzero displacement d, restoring heading. Let q_i be its phase read positions in the actual pose. At q_i+kd the required old-field contribution is determined by

```
actual required turn bit = 1_B(q_i+kd)
   XOR #{(j,s): q_j-q_i=s d,
                s>0 or (s=0 and j<i), 0<=s<=k} mod 2. (MACRO4)
```

This computes the literal first mismatch by induction up to that mismatch; it is not a finite-repeat heuristic. Prior-write changes occur at finitely many epoch thresholds.

For a finite segment a+jv, 0<=j<L, its intersection with a phase ray q+kd is empty, a singleton, or a finite arithmetic interval in k. Nonparallel directions use an exact determinant. For parallel directions solve the one-dimensional integer congruence after dividing by the greatest common divisor. In the closed stride family its epoch modulus is 1 or 2. This modulus matters: the 24-mode has axis stride 2 and the 28-mode axis stride 4. Treating every parallel intersection as a full interval would be wrong.

Split epochs into k=2z and k=2z+1. Each contribution is then an ordinary finite interval in z; each write threshold is a final interval. Sort parity-change endpoints separately for the two classes, including z=0 and the final infinite interval. This returns the exact earliest mismatch, or certifies none. All arithmetic is on integers of polynomial bit size.

Before a finite mismatch at t=nP+j, append P finite write segments for the full periods (if n>0), and j singleton writes for the partial period. These are XOR contributions; EVERY skipped write is accounted for. There are at most 2P-1 new terms. Shift the history vertex around its simple cycle by t mod P. Read the real cell and execute its actual update. If J10 is true, return the complete post-state. Otherwise the real edge must leave this simple component: the unique internal edge has the different predicted label. Continue in the lifted graph.

If no mismatch exists, the actual infinite continuation equals this cycle word. A finite field cannot realize the 24- or 28-word forever: their stabilized first-read classes require respectively two and five infinite initial black rays, as proved in the prior theorem. Hence only the standard cycle can give permanent absence. All earlier actual updates were individually checked or were legal cycle edges, so the certificate says no J10 at ANY nonnegative time, not only after its standard boundary.

A component is never revisited after exit. By (MACRO3), one call therefore uses at most five plans, at most 956 noncyclic updates, at most five cycle exits, at most 149 warm-up updates, and one final event update. A safe literal bound is 1111. The sum of visited cycle periods is at most 212, hence

```
new terms <=149+956+2*212+1=1530.
```

The 2P allocation includes a possible non-event cycle exit. No skipped write or exit update is omitted. Periodic proposals and inverse-history computations use a copy until the actual endpoint is determined.

4.4 A generic finite periodic-plan halo lemma

This lemma concerns ANY supplied periodic turn word of length P restoring heading with d!=0, from a complete finite source Y=(B,p,h). Suppose its first mismatch is at time kP+i, position e=p+q_i+kd, with phase coordinates relative to p.

Let L be the largest positive translation threshold s with q_j-q_i=s d, or zero if none. Since a period has P unit moves,

```
||q_i||_infinity<=P,   L||d||_infinity<=P.
```

If k<=L then ||e-p||_infinity<=2P. If k>=L+1, all planned previous-write parities at phases i in epochs k-1 and k have stabilized. If the stable required old-field bit is 0, mismatch means e is an original black cell. If it is 1, the preceding occurrence matched, so e-d is an original black cell. Thus in every case

```
e belongs to conv(B union {p})+[-2P,2P]^2.              (MACRO5)
```

Here B is the FULL source field before the plan, not an unknown active subset. For k>=1, any earlier phase point q_j+ell*d differs from (ell/k)(q_i+kd) by at most 2P in infinity norm. For k=0, all reads are within P of p. Therefore the entire finite planned read path through e is inside

```
conv(B union {p})+[-4P,4P]^2.                           (MACRO6)
```

Its ensuing literal update enlarges this by at most one. Unchanged source cells remain present outside the read footprint; (MACRO6) never authorizes deleting them.

For radius R=max_{b in B}||b-p||_infinity (zero for empty B), (MACRO5)'s more precise alternatives give: either k<=L<=P, or k||d||<=R+||d||+P. Therefore the first mismatch time is strictly less than

```
P(R+P+2).                                               (MACRO7)
```

These are conditional-on-first-mismatch bounds. A valid infinite plan has no finite halo bound.

4.5 All-input J10 geometry, time and bit complexity

Sum (MACRO6) over all finite periodic portions of one event call. The period sum is at most 212; all actual literal updates number at most 1111. Convex Minkowski enlargements compose, giving

```
H=4*212+1111=1959.
```

Every intermediate complete state lies in the initial complete convex hull enlarged by H; in particular its width is at most W0+2H and its radius about its current head at most that width. Applying (MACRO7) to every periodic part, using P<=104, and including all literal updates, gives post-event duration at most

```
212(W0+2H+106)+1111 =212W0+854199.                      (MACRO8)
```

After n-1 event calls width is at most W0+2H(n-1). Summing (MACRO8) gives (MACRO1), since 212H=415308. No theorem bounds n. In particular an infinite chain is not classified by the finiteness of each call.

The support of every newly appended segment is made of actual read positions. Thus all its endpoints lie in the same enlarged initial hull, even if it later cancels other terms. After n events, all coordinate and segment-length bit sizes are O(b+log(n+1)); event times also have O(b+log(n+1)) bits by (MACRO1). The number of terms is at most M+1530n. Each call performs a fixed number of ray/segment intersections, endpoint sorts and fixed-length literal or inverse operations. Exact set/list arithmetic therefore proves poly(M+n+b) bit complexity. Fixed constants may be large; this is not a benchmark or a claim polynomial in log n.

Together with the already proved H(X) iff finitely many J10 events, (MACRO1) implies that every hypothetical finite-support HC counterexample, of EITHER the nonstandard-periodic or nonperiodic type, has at least n intrinsic J10 events before every time T with (MACRO1)'s right side <=T. In particular its liminf event count divided by sqrt(T) is at least 1/sqrt(415308). This counts intrinsic events; interval-contained ten-factor counts differ only in the first nine boundary positions.

This is a lower rate requirement, not an upper event budget, a NO-HIGHWAY certificate, or a contradiction. The mass-defect lower bound and this timing bound can coexist with infinite renewal.

5 Mass and defects on arbitrary intervals

For a finite word v, put $g(v) = \#R - \#L$ and $D(v) = 3|v| - 26g(v)$. **Theorem 1 (finite-word deficit).** Every physically feasible, ten-legal finite word satisfies $D(v) \leq 322$, with no infinite-extension premise. The all-input inequality of Theorem E is denoted (MASS1) in this section.

5.1 Exact finite-word feasibility and enumeration coverage

A prescribed turn word fixes its geometric path from (0,0),N. It is physically feasible exactly when successive reads of each physical cell request opposite turns. Necessity is the flip rule. For sufficiency, put an initial black cell precisely at each visited position whose first request is L, and leave all other positions white. Induction realizes the entire word. The required seed is finite. Arbitrary poses follow by translation and quarter turns.

This elementary criterion is prior methodology, as in Gajardo's symbolic forbidden-word treatment; it is not claimed as new. It implies in particular that finite-word feasibility is the same whether initial colourings are required to be finite or are arbitrary.

Starting at the empty word, branch on BOTH turns. Reject an extension exactly if its new ten-factor is forbidden, or its current cell was previously read with the same letter. Induction on length proves complete enumeration of physically feasible ten-legal words through depth 300. Words of lengths below ten are NOT initially restricted to the standard factor table.

The Python implementation uses breadth-first cloned last-read maps. The C++ implementation instead uses depth-first reversible current-colour assignments. No seed-

size or spatial cutoff is used. Their complete sorted depth-300 edge sets and their integer potentials have identical hashes.

5.2 The finite graph certificate covers arbitrarily long finite words

Form a graph from EVERY accepted word of length 300: its 299-prefix is the source, its 299-suffix the target, and the appended letter labels the edge. There are exactly 29,334 vertices and 29,559 edges. Assign the weight

```
w(R)=-23, w(L)=29,
```

so that D(v) is the sum of its letter weights.

The certificate assigns a nonnegative integer h(u) to each vertex, with maximum 251, and verifies EVERY edge inequality

```
h(u) >= w(c)+h(v),                u --c--> v,              (MASS2)
```

and EVERY initial-vertex inequality

```
D(u)+h(u) <= 284.                                        (MASS3)
```

These are direct integer comparisons; an optimizer's output is not trusted. The potential generators are only proposal generators. Their resource caps raise an error with no mathematical conclusion. An independently supplied damaged potential is rejected without repair.

For any physically feasible ten-legal finite word z of length n>=300, all its 300-windows belong to this graph. They define a path starting with its first 299 letters u0. Summing (MASS2) over the appended letters gives

```
D(z) <= D(u0)+h(u0)-h(u_final) <= 284,
```

using (MASS3) and h>=0. This is an overapproximation argument: not every graph path need be physically realizable, but every physical word is covered. No extension to an infinite ten-legal path is assumed. There is no misuse of eventual-cycle classification for a finite segment.

For all lengths 0 through 299, exhaustive enumeration gives maximum 322. Combining the two finite kernels proves Theorem 1 for all finite lengths.

The source witness fixes SHA256 hashes for the sorted depth-300 words and the potentials in lexicographic vertex order.

The complete finite sets are reconstructed deterministically and checked against those hashes and the local inequalities. They need not be stored as a nine-megabyte enumeration dump. verification/mass/witness.json contains the immutable numerical claim, exact short witness and its complete minimal seed.

5.3 Sharpness

The word

```
LLLLRLRRRRLRLLRRLLLLRRLLRRRRLLRRLRLLRLLRLRLLRRRRLLRLRRRRLRLLRRLLLLRRLLRRRRLLRRLRL
↪ LRLLRLRLL
```

has length 90, 44 right turns and 46 left turns. It is physically feasible and all of its 81 ten-factors are allowed. Its unique minimal first-read seed has 28 black cells and is listed in verification/mass/witness.json. Literal execution leaves 26 black cells. Thus

```
3*90-26*(26-28)=322,
```

proving the additive constant in Theorem 1 and in (MASS1), with wholly contained windows, cannot be decreased.

The standard word has 104 updates and 12 more R than L. Its certified finite-support realization supplies physically feasible words W^k for every k>=1. Thus any uniform lower estimate g>=rho*n-C with rho>12/104=3/26 fails as k tends to infinity. The standard tail is an existing certified input here; this work uses that standard growth as its sharpness witness. The supplementary literal W^4 check is a control, not the justification of the universal k quantifier.

5.4 Derivation of the unrestricted inequality

In any actual finite orbit segment, every right turn increases complete black mass by one and every left turn decreases it by one. Hence

```
g(a_s...a_(t-1))=|B_t|-|B_s|.                                (MASS4)
```

Remove each letter that ends a forbidden ten-window wholly contained in the segment. These N=N10(s,t) single-letter positions separate at most N+1 consecutive remaining runs. Every ten-window wholly inside a remaining run was wholly inside the original segment and cannot be forbidden: otherwise its last position would have been removed. Each run is a genuine physical subword and is therefore covered by Theorem 1. An empty run has deficit zero and may harmlessly be charged the same upper bound.

Each removed letter has deficit at most 29. Additivity of D, Theorem 1 and (MASS4) yield

```
D(total) <= 322(N+1)+29N = 351N+322,
```

proving (MASS1). This does not require disjoint forbidden windows. Removing their ending letters handles overlaps exactly.

5.5 Density consequences

Taking $s = 0$ and dividing by t gives two immediate consequences: if $|B_t| = o(t)$, then $\liminf N_{10}(0,t)/t \geq 1/117$; if $N_{10}(0,t) = o(t)$, then $\liminf |B_t|/t \geq 3/26$. These quantify the cost of maintaining low mass or a nearly legal stream.

6 Reproducibility and conclusion

The source package provides two complete finite kernels. The attraction kernel regenerates the 6,605 physical length-150 words, the quotient of the nonsimple component, and the exact path and supply bounds. The mass kernel regenerates all 29,559 length-300 edges, checks every integer potential inequality, and replays the equality

witness. Python and C++ implementations are compared, and damaged certificates are rejected. The event lift additionally checks all 7763 edges and component-rank maxima 5, 956 and 212. The all-parameter foldback exclusion, seed induction and spiral separation are proved in the text; finite geometric checks verify their stated numerical premises.

Run `python3 factor_language/reproduce.py` from the source root. Retained source hashes and a fresh per-stage report distinguish the imported evidence from the checks performed for this manuscript. These are analytic proofs with exact computation. The older immediate-standardness thresholds concern a different conclusion and are not premises of this paper.

The two proof routes expose complementary structure: a legal infinite future has rigid eventual motion, while an arbitrary finite interval must pay for its mass deficit with local violations. Their combination connects a local signature of P104 with both a decidable event and a quantitative all-input law.

A Exact words

The machine-readable words are in `verification/ten/certificates/witness.json`; line breaks below are not letters. All geometric statements use north at the origin unless a phase placement is explicit.

A.0.1 W (length 104)

RRLLLLRLRRRRLRRRRLLLLRLRRRRLRLLRRLLLLRRLLRRRRLLRRLRL
LRLLRLRLLRRRRLLRLLRRRRLLRRRRLLRLRRRRLRLLLLRRRRLRRLRR

A.0.2 U (length 28)

RRLRLLLLRRRRLRRLRRRRLLLLRLRR

A.0.3 T (length 24)

RRRRLLRLLRRRRLLRRRRLLRLL

A.0.4 A (length 28)

LLRRRRLRRLRRRRLLLLRLRRRRLRLL

A.0.5 B (length 102)

RRLLLLRRLLRRRRLLRRLRLLRLLRLRLLRRRRLLRLRRRRLRLLLLRRRR
LRRLRRRRLLLLRLRRRRLRLLLLRRRRLRRLRRRRLLLLRLRRRRLRLL

A.0.6 C (length 89)

RRLRRRRLLLLRLRRRRLRRLRRRRLLLLRLRRRRLRLLLLRRRRLRRLRRR
RLLLLRLRRRRLRLLLLRRRRLRRLRRRRLLLLRLRR

A.0.7 Z (length 104)

```
RRRRLLRLRRRRLRLLLLRRRRLRRLRRRRLLLLRLRRRRLRRRRLLLLRLR
RRRLRLLRRLLLLRRLLRRRRLLRRLRLLRLLRLRLLRRRRLLRLLRRRRLL
```

B First-black supply table for the 24-mode

The two rays are each advanced by 24. The finite times and the rays are disjoint as read events and correspond to distinct first-read cells.

Phase	Finite first-L times	Ray starts	Maximum $L - 12b_j(L)$
0	5,7	8,23	5
1	3,4,6	7,22	3
2	2,3,5	6,21	2
3	1,2,4	5,20	1
4	0,1,3	4,19	0
5	0,2	3,18	0
6	1	2,17	1
7	0,7	1,16	0
8	6	0,15	0
9	5	14,23	5
10	3,4	13,22	3
11	2,3,11	12,21	2
12	1,2,10,19	11,20	1
13	0,1,7,9,18	10,19	0
14	0,6,8,17	9,18	0
15	5,7,16	8,17	5
16	3,4,6,15	7,16	3
17	2,3,5,14	6,15	2
18	1,2,4,13	5,14	1
19	0,1,3,12	4,13	0
20	0,2,11	3,12	0
21	1,10	2,11	1
22	0,7,9	1,10	0
23	6,8	0,9	0

C References

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2. Anahí Gajardo, Victor Lutfalla and Michaël Rao. *Ants on the highway* (2024). arXiv:2409.10124v1.
3. Atharva Jillhewar. *Finite-Support Periodic Highways of Langton's Ant: Necessary Conditions, Transverse Exclusions, and Exact Search* (2026). doi:10.21203/rs.3.rs-10516269/v1. Theorem 3.1 and Remark 3.2 supply full-drift visit chronology and seed reconstruction.

4. Hao Ke. *An Ant on the Clock: Exponential Escape Bounds for Langton's Ant,* version 0.4.0 (2026). doi:10.5281/zenodo.23402379.